

Probability Theory - Winter term 2025/26

Personal Solution Set 4

TU Dortmund

Problem 1

1. Let X be a random variable with $P(X > 0) > 0$. Prove or disprove that there is a constant $\delta > 0$ such that $P(X \geq \delta) > 0$ holds.
2. Let $f : (\Omega_1, \mathcal{A}_1) \rightarrow (\Omega_2, \mathcal{A}_2)$ be $\mathcal{A}_1 - \mathcal{A}_2$ measurable and $g : (\Omega_2, \mathcal{A}_2) \rightarrow (\Omega_3, \mathcal{A}_3)$ be $\mathcal{A}_2 - \mathcal{A}_3$ measurable. Show that $g \circ f$ is $\mathcal{A}_1 - \mathcal{A}_3$ measurable.
3. Let $X : [0, 1] \rightarrow \Omega = \{0, 1, 2\}$ be defined by $X(\omega) = 1$ if $\omega \in [0, 1/3]$, 0 if $\omega = 0$, and 2 else. Show X is a random variable on $([0, 1], \mathcal{B}([0, 1]), P)$ and state its probability distribution under $P = \lambda_{[0,1]}$.

Solution

1. Increasing sequence of events $A_n := \{\omega \in \Omega \mid X(\omega) > 1/n\}$. The sequence satisfies $A_n \subseteq A_{n+1}$. The union of these sets is exactly the event $A := \{X > 0\}$. Using continuity from below for the probability measure P :

$$\lim_{n \rightarrow \infty} P(A_n) = P(A) > 0.$$

Because the limit is strictly greater than 0, there must exist some $n_0 \in \mathbb{N}$ where $P(A_{n_0}) > 0$. Choose $\delta = 1/n_0$, that shows $P(X \geq \delta) \geq P(X > 1/n_0) > 0$ the statement is proven.

2. Let $B \in \mathcal{A}_3$. To prove measurability of the composition $g \circ f$, I need to show $(g \circ f)^{-1}(B) \in \mathcal{A}_1$. Using props of pre-images:

$$(g \circ f)^{-1}(B) = f^{-1}(g^{-1}(B)).$$

Because g is $\mathcal{A}_2 - \mathcal{A}_3$ measurable, $g^{-1}(B) \in \mathcal{A}_2$. Because f is $\mathcal{A}_1 - \mathcal{A}_2$ measurable, taking the pre-image of that resulting set under f ensures $f^{-1}(g^{-1}(B)) \in \mathcal{A}_1$. Therefore, $g \circ f$ is measurable.

3. Verifying measurability of X by checking the pre-images of all elements in $\mathcal{P}(\Omega)$:

- $X^{-1}(\emptyset) = \emptyset \in \mathcal{B}([0, 1])$
- $X^{-1}(\{0\}) = \{0\} \in \mathcal{B}([0, 1])$
- $X^{-1}(\{1\}) = (0, 1/3] \in \mathcal{B}([0, 1])$
- $X^{-1}(\{2\}) = (1/3, 1] \in \mathcal{B}([0, 1])$

Unions of these sets also result in valid Borel sets (e.g., $X^{-1}(\{0, 1\}) = [0, 1/3]$). Thus, X is a measurable function and a valid random variable. To find the distribution under the Lebesgue measure λ , $P^X(B) = \lambda(X^{-1}(B))$ for the singletons:

$$\begin{aligned} P^X(\{0\}) &= \lambda(\{0\}) = 0 \\ P^X(\{1\}) &= \lambda((0, 1/3]) = 1/3 \\ P^X(\{2\}) &= \lambda((1/3, 1]) = 2/3 \end{aligned}$$

Problem 2

The probability density function of a $\text{Gamma}(\alpha, \beta)$ distribution is $f(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$ for $x > 0$.

- Show directly that the gamma pdf is valid.
- Calculate the expected value of X .
- Derive the distribution of $Y = 1/X$ and its expected value.
- How would you simulate an exponentially distributed random variable with parameter β ?

Solution

- The function $f(x)$ is clearly non-negative for $x > 0$. To verify it integrates to 1, here use the substitution $u = \beta x$, $du = \beta dx$:

$$\int_0^\infty \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x} dx = \frac{\beta^\alpha}{\Gamma(\alpha)} \int_0^\infty \left(\frac{u}{\beta}\right)^{\alpha-1} e^{-u} \frac{1}{\beta} du = \frac{\beta^\alpha}{\Gamma(\alpha)} \frac{\Gamma(\alpha)}{\beta^\alpha} = 1.$$

- Using the definition of expectation and the same substitution $u = \beta x$:

$$E(X) = \int_0^\infty x \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x} dx = \frac{\beta^\alpha}{\Gamma(\alpha)} \frac{1}{\beta^{\alpha+1}} \int_0^\infty u^\alpha e^{-u} du = \frac{\Gamma(\alpha+1)}{\beta \Gamma(\alpha)} = \frac{\alpha}{\beta}.$$

- density transformation formula with $y = \phi(x) = 1/x$ and its inverse $\phi^{-1}(y) = 1/y$. The derivative is $|d/dy(1/y)| = 1/y^2$.

$$f_Y(y) = f_X(1/y) \left| -\frac{1}{y^2} \right| = \frac{\beta^\alpha}{\Gamma(\alpha)} \left(\frac{1}{y}\right)^{\alpha-1} e^{-\beta/y} \frac{1}{y^2} = \frac{\beta^\alpha}{\Gamma(\alpha)} \left(\frac{1}{y}\right)^{\alpha+1} e^{-\beta/y}.$$

This is the density of an Inverse-Gamma distribution. The expected value is:

$$E(Y) = \int_0^\infty y f_Y(y) dy = \int_0^\infty \frac{\beta^\alpha}{\Gamma(\alpha)} y^{-\alpha} e^{-\beta/y} dy.$$

Substituting $u = \beta/y$ ($dy = -\beta/u^2 du$):

$$E(Y) = \frac{1}{\Gamma(\alpha)} \int_0^\infty u^{\alpha-2} \beta e^{-u} du = \frac{\beta \Gamma(\alpha-1)}{\Gamma(\alpha)} = \frac{\beta}{\alpha-1} \quad \text{for } \alpha > 1.$$

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Problem 3

Let X be a random variable such that $E[|X|^l] < \infty$ for some $l \in \mathbb{N}$. Show that:

- a) $E[|X|^k] < \infty$ for all $k \leq l$.
- b) $E[|X - c|^l] < \infty$ for all $c \in \mathbb{R}$.

Solution

- a) For $k \leq l$, bound $|X|^k$ from above:

$$|X|^k \leq 1 + |X|^l.$$

By the monotonicity and linearity of the expectation:

$$E(|X|^k) \leq E(1 + |X|^l) = 1 + E(|X|^l) < \infty.$$

- b) Using the binomial theorem and the triangle inequality:

$$|X - c|^l = \left| \sum_{k=0}^l \binom{l}{k} X^k (-c)^{l-k} \right| \leq \sum_{k=0}^l \binom{l}{k} |X|^k |c|^{l-k}.$$

Taking the expectation on both sides:

$$E(|X - c|^l) \leq \sum_{k=0}^l \binom{l}{k} |c|^{l-k} E(|X|^k).$$

Since $E(|X|^k) < \infty$ for all $k \leq l$ (from part a), the right-hand side is a finite sum of finite terms. Therefore, $E(|X - c|^l) < \infty$.